

MAUI ·

■■■■■2026■5■24■ — 5■26■■■■1-3■■■

Page 10 of 10

- [illegible]

Page 10 of 10

Icon	Year	Platform	Framework
MAUI	2023	.NET MAUI	MAUI API
Android	2020	Android Java + SpringBoot	Android + API
Cordova	—	Cordova/Android	UI/UX API
d2030	2030	WXXML/WXSS	
Flutter	2023	Flutter Dart	Flutter AI

100%

Page 1 of 1

1. ■■■■■■

- ■■■■■■ .NET MAUI ■■■■■■
- ■■■■■ C# + XAML ■■ MAUI ■■■■■■■■■■■■■■■■■■■ API ■■■
- ■■■■■ Android ■ iOS ■ Windows ■■■ MacCatalyst ■ Tizen ■

2. ■■■■■■MAUI■■■

[illegible]

3. 项目背景

- 项目背景
- 项目背景 API
- 项目背景

4. 项目背景

- 项目背景 QQ
- 项目背景
- 项目背景 Gitee
- 项目背景 20 项目 15 项目

项目背景

1. 项目背景

2. MAUI 项目背景

1. 项目背景 / 项目 / 项目 / 项目
2. 项目背景
3. 项目背景 / 项目 / 项目
4. AI 项目 DeepSeek V4 Pro + 项目 + 项目
5. 项目背景 AI
6. 项目背景
7. 项目
8. 项目背景 / 项目 / 项目 / 项目
9. 项目背景
10. 项目背景
11. 项目背景
12. 项目背景
13. 项目
14. 项目背景
15. 项目
16. 项目

项目背景

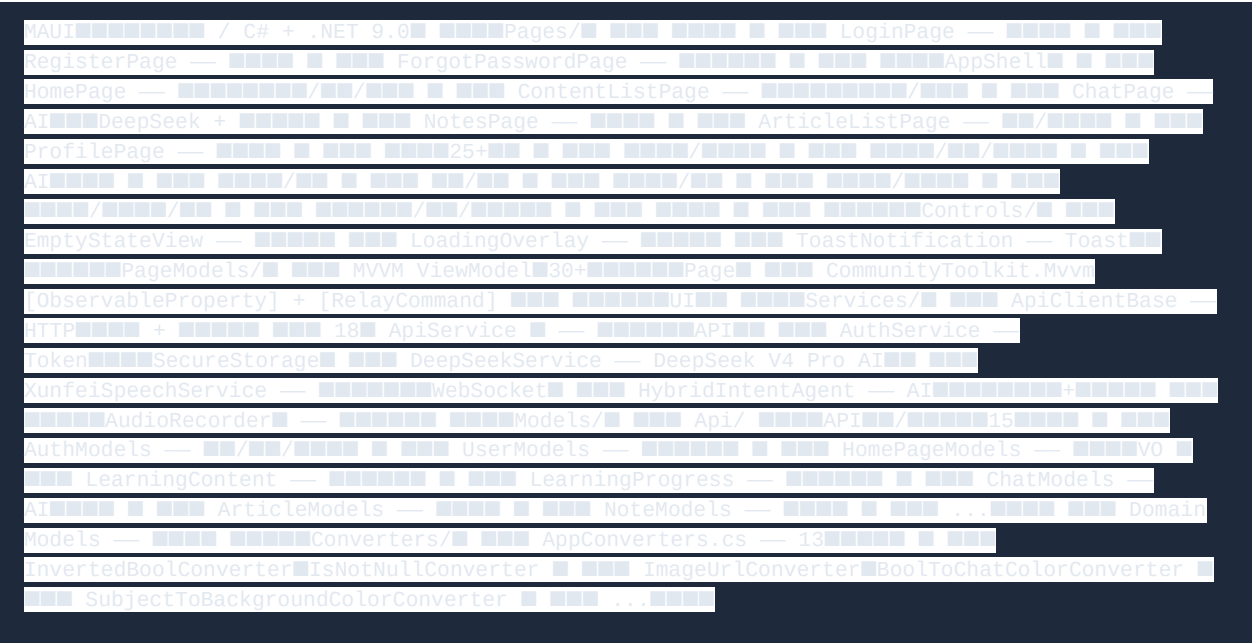
项目背景

1. 项目背景

1-1 MAUI 项目背景 → 项目 → 项目 → 项目

1-2 MAUI

1-3 MAUI ApiClientBase + AuthHandler + Service



1-4 LoginPage → AuthApiService → SecureStorage

1-5 AI → ChatPage → DeepSeekService → API →

1-6

2. MAUI

	C#	.NET 9.0
UI	MAUI + XAML	UI
MVVM	CommunityToolkit.Mvvm 8.3.2	
UI	CommunityToolkit.Maui 11.1.1	UI
	Syncfusion.Maui.Toolkit 1.0.6	
	HttpClient	
	Microsoft.Data.Sqlite.Core 8.0.8	SQLite
	NAudio 2.2.1 Windows	
AI	DeepSeek API deepseek-v4-pro	HTTP POST
	WebSocket API	
	SecureStorage	Token

3. API

MAUI API 20+

/api/auth/login	POST		
/api/auth/register	POST		
/api/auth/forgot-password	POST		
/api/user/info	GET/PUT		
/api/home/dashboard	GET		
/api/contents	GET		
/api/learning-progress	GET/PUT		
/api/ai/chat	POST	AI	
/api/notes	GET/POST/PUT/DELETE	CRUD	
/api/articles	GET/POST		
/api/checkin	POST		
/api/reminders	GET/POST/PUT/DELETE		
/api/parent/rules	GET/POST/PUT/DELETE		
/api/admin/contents	GET/POST/PUT/DELETE		
/api/upload/image	POST		
/api/announcements	GET		

Knife4j OpenAPI

API `Result<T>` {code, message, data}

Knife4j 16 public_OpenAPI.json

4.

MAUI API MySQL

- SecureStorage JWT Token 256 ID

- SQLite 13

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1 MAUI

- [illegible]

2 API

- ```
- ■■■■■■■■■■■■■■■■■■■■■■API■■■■■■■■
- ■■■■■■■■■Postman■■■ /api/auth/login ■■■■■■■■■■■■■■■■■■■■■ user1 / 123456■■■■■■■■■■

code/message/data/token ■■■

- ■■■■■■■■■■■■■■■■■■■ code: 0, data: {token, user}■■■■■■■■ ~200ms■
- ■■■■■■■■■■■■■ http://47.110.3.200:8081
```

### 3 CommunityToolkit.Mvvm

- ```
- ■■■■■■■■■■MVVM■■■■ [ObservableProperty] ■ [RelayCommand] ■■■■■■■■■■
- ■■■■■■■■■■ViewModel■■■■ [ObservableProperty] private string _testProp■■■■■■■■■■■
TestProp ■■■■■■ OnTestPropChanged ■■■■
- ■■■■■■■■ ■■
```

4 DeepSeek API

- [illegible]

2. ■■■■■■

□□□□	□□□□	□□□□	□□	□□□□	□□□□
AI□□□□	DeepSeek API	AI□□□□□□□□□□	Token□□□□	□□	□□
□□□□	□□WebSocket API	□□□□□□□□	□□□□□□	□□	□□
□□□□	QQ□□SMTP	□□□□□□	□□	□□	□□
API□□	Knife4j	□□□□□□□□	□□□□	□□	□□

3. MAUI

- < 3 Windows/Android
- < 300
- < 2 AI 10
- < 200 MB
- < 80 MB

4.

	API		ApiClientBase
	DeepSeek API		<50ms+
	30+15		→ → → AI



1.

IDE	Visual Studio 2022	
SDK	.NET 9.0	
MAUI	maui workload	
	Windows	
	Git feature/maui	

1-7 Visual Studio 2022 ~120MB CPU 2-5%

2. MAUI

	MauiApp.sln
	NET 9.0
	Android 34.0.0
	Android 34.0.0
	Android 34.0.0
	Android 34.0.0
	Android 34.0.0
	Android 34.0.0

3. NuGet

Package	Version	Platform
CommunityToolkit.Mvvm	8.3.2	MVVM
CommunityToolkit.Maui	11.1.1	MAUI UI
Syncfusion.Maui.Toolkit	1.0.6	Syncfusion/MAUI
Microsoft.Data.Sqlite.Core	8.0.8	SQLite
NAudio	2.2.1	Windows

4.

```
// ApiClientBase.cs — HTTP
public class ApiClientBase {
    protected readonly HttpClient _httpClient;
    protected readonly string _baseUrl;

    public ApiClientBase(HttpClient httpClient) {
        _httpClient = httpClient;
        _baseUrl = "http://47.110.3.200:8081";
    }

    protected async Task<ApiResponse<T>> GetAsync<T>(string endpoint) {
        var response = await _httpClient.GetAsync($"{_baseUrl}{endpoint}");
        response.EnsureSuccessStatusCode();
        var json = await response.Content.ReadAsStringAsync();
        return JsonSerializer.Deserialize<ApiResponse<T>>(json, new JsonSerializerOptions {
            PropertyNameCaseInsensitive = true });
    }

    protected async Task<ApiResponse<T>> PostAsync<T>(string endpoint, object body) {
        var content = new StringContent(JsonSerializer.Serialize(body), Encoding.UTF8, "application/json");
        var response = await _httpClient.PostAsync($"{_baseUrl}{endpoint}", content);
        response.EnsureSuccessStatusCode();
        var json = await response.Content.ReadAsStringAsync();
        return JsonSerializer.Deserialize<ApiResponse<T>>(json, new JsonSerializerOptions {
            PropertyNameCaseInsensitive = true });
    }

    // PUT / DELETE
    protected async Task<ApiResponse<T>> PutAsync<T>(string endpoint, object body) {
        var content = new StringContent(JsonSerializer.Serialize(body), Encoding.UTF8, "application/json");
        var response = await _httpClient.PutAsync($"{_baseUrl}{endpoint}", content);
        response.EnsureSuccessStatusCode();
        var json = await response.Content.ReadAsStringAsync();
        return JsonSerializer.Deserialize<ApiResponse<T>>(json, new JsonSerializerOptions {
            PropertyNameCaseInsensitive = true });
    }

    protected async Task<ApiResponse<T>> DeleteAsync<T>(string endpoint) {
        var response = await _httpClient.DeleteAsync($"{_baseUrl}{endpoint}");
        response.EnsureSuccessStatusCode();
        var json = await response.Content.ReadAsStringAsync();
        return JsonSerializer.Deserialize<ApiResponse<T>>(json, new JsonSerializerOptions {
            PropertyNameCaseInsensitive = true });
    }
}

// AuthHandler.cs — Bearer Token
public class AuthHandler : DelegatingHandler {
    private readonly SecureStorage _secureStorage;

    public AuthHandler(SecureStorage secureStorage) {
        _secureStorage = secureStorage;
    }

    protected override async Task<HttpResponseMessage> SendAsync(HttpRequestMessage request) {
        var token = await _secureStorage.GetAsync("token");
        request.Headers.Authorization = new AuthenticationHeaderValue("bearer", token);
        return await base.SendAsync(request);
    }
}

// MauiProgram.cs — DI
public MauiProgram MauiProgram {
    get {
        return new MauiProgram(
            builder => {
                builder.Services.AddSingleton<AuthHandler>();
                builder.Services.AddHttpClient<ApiClientBase>(...).AddHttpMessageHandler<AuthHandler>();
            }
        );
    }
}

// Service/ViewModel/Page
public class Service {
    // ...
}

public class ViewModel {
    // ...
}

public class Page {
    // ...
}
```

5.

- PascalCase / camelCase C#
- XML
- Pages/ PageModels/ Services/ Models/ Converters/ Controls/ Platforms/
- MVVM Page + PageModel ViewModel Service

File	Content
AuthHandler.cs	AuthHandler.cs
ApiClientBase.cs	ApiClientBase.cs
MauiProgram.cs	MauiProgram.cs
Service.cs	Service.cs
ViewModel.cs	ViewModel.cs
Page.cs	Page.cs

- PR1
- Conventional Commits feat/fix/refa
- main + develop + feature/[]-[]
- +

